SHRI MADHWA VADIRAJA INSTITUTE OF TECHNOLOGY AND MANAGEMENT

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Big Data Analytics (BCS714D)

Assignment Report On

Food Ordering System

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Food Ordering System

Introduction

The Food Ordering System is a software application designed to simplify the process of ordering food from restaurants using digital platforms such as websites or mobile applications. With the rapid growth of internet usage and smartphones, traditional methods of ordering food manually or through phone calls have become inefficient and time-consuming. The Food Ordering System provides a modern solution by enabling customers to browse menus, place orders, make payments, and track deliveries online.

This system allows users to select food items from a digital menu displayed with images, prices, and descriptions. Based on the images provided, the system includes screens such as user login, food menu listing, cart page, order summary, payment page, and order confirmation page. These screens together create a smooth workflow for the customer, starting from login and ending with successful order placement.

The Food Ordering System reduces manual work for restaurant staff and minimizes errors that occur due to miscommunication. It also improves customer satisfaction by offering convenience, speed, and transparency. Customers can order food anytime and from anywhere without visiting the restaurant physically.

For restaurant owners, this system helps manage orders efficiently, monitor customer demand, and improve service quality. The system stores all order details securely in a database, which helps in maintaining records and generating reports. Additionally, online payment options make transactions faster and safer.

In today's competitive food industry, having an online food ordering system is essential. It helps restaurants reach more customers, increase sales, and build a strong online presence. Thus, the Food Ordering System plays an important role in transforming traditional food services into smart digital solutions.

System Overview

The Food Ordering System is developed using a client-server model where the user interacts with the system through a graphical interface, and all processing is handled by the backend server. The system mainly consists of three components: customer interface, admin interface, and backend database.

The customer interface is the front-end part of the system. From the images, it is clear that users can log in, view available food items, add items to the cart, place orders, and make payments. The interface is designed to be user-friendly so that even non-technical users can easily place orders. Food items are displayed with images and prices, which helps users make quick decisions.The admin or restaurant side of the system is responsible for managing food items, prices, and orders. Restaurant staff can view incoming orders, update order status, and ensure timely preparation and delivery. This module helps in smooth coordination between the restaurant and customers.

The backend system handles business logic such as order processing, user authentication, payment verification, and data storage. A database is used to store user details, food items, order history, and payment records. The backend communicates with the frontend using APIs to exchange data securely.

Overall, the system ensures smooth data flow between users, restaurants, and the server. Each component works together to provide a reliable and efficient food ordering experience.

Functional Modules

The Food Ordering System is divided into several functional modules to perform specific tasks efficiently.

The User Registration and Login Module allows users to create an account and log in securely. Authentication ensures that only registered users can place orders. The login screen shown in the images verifies user credentials before allowing access.

The Food Menu Module displays available food items with images, names, prices, and descriptions. Users can browse different categories and select items

of their choice. This module enhances user experience through visual presentation.

The Cart Management Module enables users to add selected food items to the cart. Users can increase or decrease quantities, remove items, and view the total cost. This module ensures accurate billing before order placement.

The Order Placement Module allows users to review their selected items and confirm the order. Delivery address and contact details are verified before final submission.

The Payment Module supports online payment options such as UPI, cards, or cash on delivery. Secure payment processing ensures safe transactions.

The Order Confirmation and Tracking Module provides confirmation after successful payment. Users can track order status from preparation to delivery.

Each module works independently yet is interconnected, ensuring smooth system operation.

System Architecture and Technologies Used

The Food Ordering System follows a three-tier architecture consisting of the presentation layer, application layer, and data layer.

The presentation layer includes the user interface seen in the images. It is developed using technologies such as HTML, CSS, JavaScript, or React. This layer handles user interactions and displays information.

The application layer contains the business logic. It processes user requests, validates data, manages orders, and handles payments. Backend technologies such as Node.js, Java, or Python are commonly used.

The data layer consists of the database used to store user data, food details, and orders. Databases like MongoDB or MySQL ensure secure and efficient data storage.

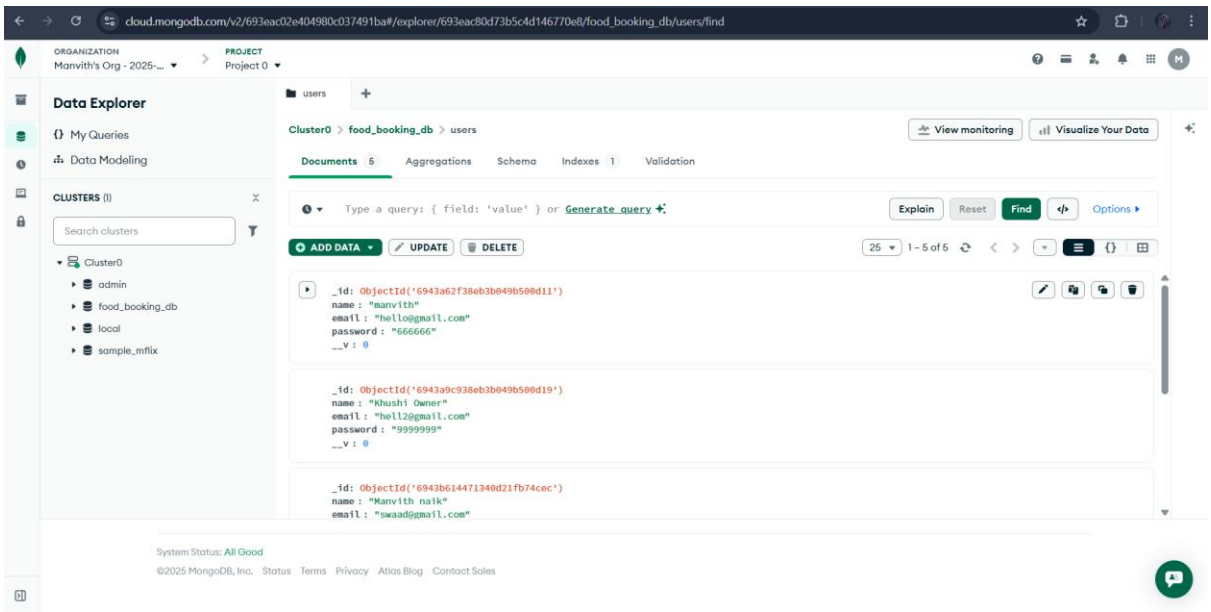
APIs are used to connect the frontend and backend, ensuring smooth communication.

Payment gateways are integrated to support online transactions.

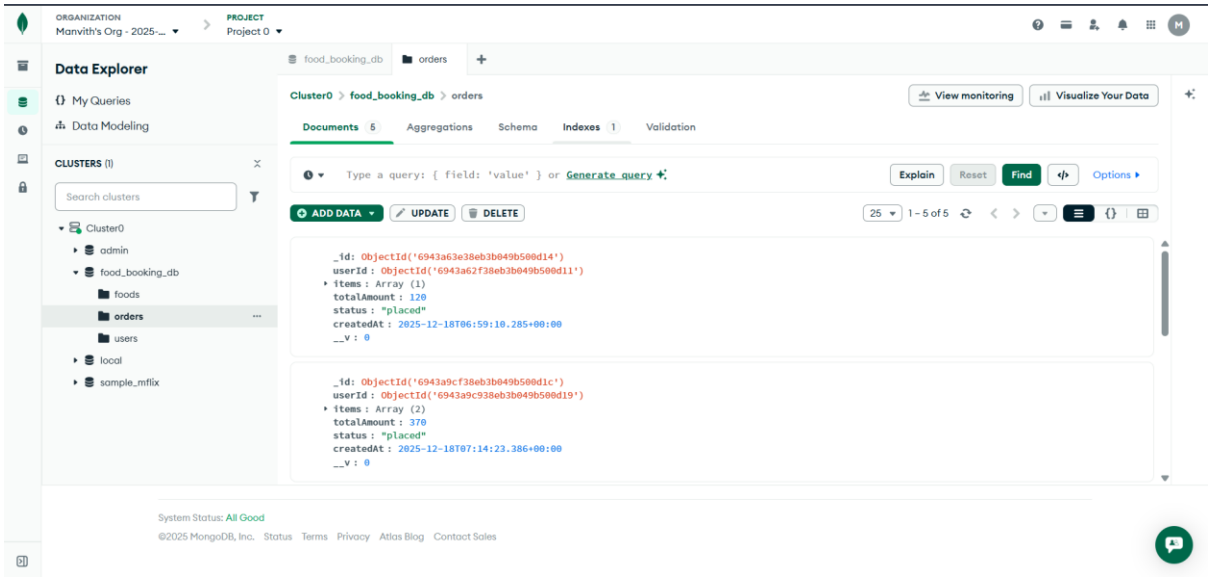
This architecture ensures scalability, security, and reliability of the system.

MongoDB

users



orders



Project link :

https://food-booking-system-five.vercel.app/

Advantages, Limitations, and Future Scope

The Food Ordering System offers many advantages. It saves time for customers, reduces manual errors, and provides 24/7 availability. Restaurants can manage orders efficiently and improve customer satisfaction. Digital payments ensure faster transactions.

However, the system also has limitations. It depends on internet connectivity and may face technical issues such as payment failures or server downtime. Initial development and maintenance costs can be high.

In the future, the system can be enhanced by adding features such as

AI-based food recommendations, real-time delivery tracking, offers and discounts, chat support, and multi-language support. Mobile app integration can further improve accessibility.

Conclusion

The Food Ordering System is an efficient digital solution that modernizes the food ordering process. By referring to the system screens shown in the images, it is evident that the application provides a smooth, secure, and user-friendly experience. This system benefits both customers and restaurants and has great potential for future improvements.